

Supplementary Table 1

Name	Sequence and modification	Vendor
FC_SC_01_NH2	/5AmMC6/ATGAGGTGTATGTGTAGAGTGATGGATGTAGT/3AmMO/	IDT
FC_SC_02_NH2	/5AmMC6/AGGATGAGTGAGAGTGAGATGAGAGTAGATGT/3AmMO/	IDT
FC_St_02_NH2	/5AmMC6/CACTCTCACACCTCATACATCTACCATCACTC/3AmMO/	IDT
FC_SC_01_ATTO488	/5ATTO488N/ATGAGGTGTATGTGTAGAGTGATGGATGTAGT/3ATTO488N/	IDT
FC_SC_02_ATTO488	/5ATTO488N/AGGATGAGTGAGAGTGAGATGAGAGTAGATGT/3ATTO488N/	IDT
FC_St_02_ATTO488	/5ATTO488N/CACTCTCACACCTCATACATCTACCATCACTC/3ATTO488N/	IDT
FC_SC_01_ATTO565	/5ATTO565N/ATGAGGTGTATGTGTAGAGTGATGGATGTAGT/3ATTO565N/	IDT
FC_SC_02_ATTO565	/5ATTO565N/AGGATGAGTGAGAGTGAGATGAGAGTAGATGT/3ATTO565N/	IDT
FC_St_02_ATTO565	/5ATTO565N/CACTCTCACACCTCATACATCTACCATCACTC/3ATTO565N/	IDT
FC_SC_01_ATTO647N	/5ATTO647NN/ATGAGGTGTATGTGTAGAGTGATGGATGTAGT/3ATTO647NN/	IDT
FC_SC_02_ATTO647N	/5ATTO647NN/AGGATGAGTGAGAGTGAGATGAGAGTAGATGT/3ATTO647NN/	IDT
FC_St_02_ATTO647N	/5ATTO647NN/CACTCTCACACCTCATACATCTACCATCACTC/3ATTO647NN/	IDT
FC_SC_01_Cy3	/5Cy3/ATGAGGTGTATGTGTAGAGTGATGGATGTAGT/3Cy3Sp/	IDT
FC_SC_02_Cy3	/5Cy3/AGGATGAGTGAGAGTGAGATGAGAGTAGATGT/3Cy3Sp/	IDT
FC_St_02_Cy3	/5Cy3/CACTCTCACACCTCATACATCTACCATCACTC/3Cy3Sp/	IDT
FC_SC_01_Cy3N	/5Cy3N/ATGAGGTGTATGTGTAGAGTGATGGATGTAGT/3Cy3N/	IDT
FC_SC_02_Cy3N	/5Cy3N/AGGATGAGTGAGAGTGAGATGAGAGTAGATGT/3Cy3N/	IDT
FC_St_02_Cy3N	/5Cy3N/CACTCTCACACCTCATACATCTACCATCACTC/3Cy3N/	IDT
FC_SC_01_Cy5	/5Cy5/ATGAGGTGTATGTGTAGAGTGATGGATGTAGT/3Cy5Sp/	IDT
FC_SC_02_Cy5	/5Cy5/AGGATGAGTGAGAGTGAGATGAGAGTAGATGT/3Cy5Sp/	IDT
FC_St_02_Cy5	/5Cy5/CACTCTCACACCTCATACATCTACCATCACTC/3Cy5Sp/	IDT
FC_St_01_Biotin	/5Biosg/TACACATACTCATCCTACTACATCTCTCATCT	IDT
FC_St_01_HALO	Halotag Ligand (O2) TACACATACTCATCCTACTACATCTCTCATCT	Biomers
dsDNA_Biotin	/5Biosg/AAGAGTGATGGTAGATGTATGAGGTGTGAGAGTG	IDT
SD_SC_01_UN	TTATGAGGTGTATGTGTAGAGTGATGGATGTAGTTT	IDT
SD_SC_02_UN	TTAGGATGAGTGAGAGTGAGATGAGAGTAGATGTTT	IDT
SD_St_02_UN	TTCACCTCTCACACCTCATACATCTACCATCACTCTT	IDT
SD_St_02_ATTO488	/5ATTO488N/CACTCTCACACCTCATACATCTACCATCACTCTT	IDT
SD_St_02_ATTO565	/5ATTO565N/CACTCTCACACCTCATACATCTACCATCACTCTT	IDT

SD_St_02_ATTO647N	/5ATTO647NN/CACTCTCACACCTCATACATCTACCATCACTCTT	IDT
SD_St_02_Cy3	/5Cy3/CACTCTCACACCTCATACATCTACCATCACTCTT	IDT
SD_St_02_Cy5	/5Cy5/CACTCTCACACCTCATACATCTACCATCACTCTT	IDT
SD_St_02_5pr_Cy3N	/5Cy3N/CACTCTCACACCTCATACATCTACCATCACTCTT	IDT
SD_St_02_3pr_Cy3N	TTCACCTCTCACACCTCATACATCTACCATCACTC/3Cy3N/	IDT
SD_SC_01_5pr_Cy3N	/5Cy3N/ATGAGGTGTATGTGTAGAGTGATGGATGTAGTTT	IDT
SD_SC_01_3pr_Cy3N	TTATGAGGTGTATGTGTAGAGTGATGGATGTAGT/3Cy3N/	IDT
SD_SC_02_5pr_Cy3N	/5Cy3N/AGGATGAGTGAGAGTGAGATGAGAGTAGATGTTT	IDT
SD_SC_02_3pr_Cy3N	TTAGGATGAGTGAGAGTGAGATGAGAGTAGATGT/3Cy3N/	IDT
	Oligos for Compact Cube below	
LC_SC_01	TTGAAAATTATCTCGATAAGCAGAAGGACCTGTATAACTGGCAAGAGACAAGGCCGCTTCAGAA	IDT
LC_SC_02	AGGATAGCCGGACCGTATTAATGCCGCGCCAACGGTTTCCCGGACCTAGTGTCTATCAAGTCTA	IDT
LC_SC_03	TTCTATGAAACCATTCTCGGGTCGAGCGGGTCACTGTTGTGACCTACGAGAAGCGTATAGATGT	IDT
LC_SC_04	TCCGCGCGAATAGCTCACAGGCGAACTACGTATGAATTGGTTTAAACGCTCCTCGGGAATTAAT	IDT
LC_SC_05	ACGACAGGTGGCAAACCACTCCGATGTCAGCGCCGCATACCCATTCACTGTGAATTTCCACAC	IDT
LC_SC_06	CGAGGATTCGCAGGTCCATGGGATTACCAAGCTCGTATACACCCTGATTCTCCATGGCAGCGC	IDT
LC_SC_07	GTAAGTTGAAGTAGGAAGCTTTTCTAGCCATAGCATCGACACTACGACCTGCTTTTCGACACA	IDT
LC_SC_08	GGACTGCATTCTGGACAGTAAGTGCATTAAGTACGTGCTCCCAACATAAGTGACGTCTCAGCA	IDT
LC_St_01	TTTGCTGAGGTGGAATTTT	IDT
LC_St_02	TTCCGGGAAACCGTTGGCCCTTCTGCTCGCCTGTCGTAGGTCGGGT	IDT
LC_St_03	GGTTCTGCGAATCCTCGGTGACGTCACTTCCTACTTCAACTTACTT	IDT
LC_St_04	GTATGCGGCGCTCAGTTACTTCGTAGTGTGATGCTTT	IDT
LC_St_05_Biotin	/5Biosg/TTCCATGGACTCATAGAATT	IDT
LC_St_06	TTGAGAATCAACAACAGTTT	IDT
LC_St_07_Cy3	TTACATCTATCACTAGGTTT/Cy3	IDT
LC_St_08	TTTACGTAGTTTATCGAGTT	IDT
LC_St_09	TTCCGGCTATCCTTTCTGAAGCGGCCCGAGGAGGAAT	IDT
LC_St_10	GTATACGAGCTTAAAAAGCTTATGTTGGGAGCACGTTT	IDT
LC_St_11	TTTGCCAGTTATACAGGTGCGGCATTGACCCGACGTTTAAAAATGG	IDT
LC_St_12	TTAGGTGGTTTCGCGCGGATT	IDT
LC_St_13	TTAGTTAATGGACATCGGTT	IDT
LC_St_14	TTATAATTTTCAATAGACTTGATAGAACGCTTCTGAGCTATTTGCC	IDT

LC_St_15	TTGACCCGCTAATACGGTTT	IDT
LC_St_16	ACCTGTCGTGCGAAAGCAGGGTCCAGAATGCAGTCCTT	IDT
LC_St_17	TTATGGCTAGGGTGAATCTT	IDT
LC_St_18	TTCACAGTGACCAATTCATT	IDT
LC_St_19	TTTGTGTCGACTGCCATGTT	IDT
LC_St_20	TTATTAATTCTTGCTCTTT	IDT
LC_St_03_Cy3	GGTTCTGCGAATCCTCGGTGACGTCACCTTCTACTTCAACTTACTT/Cy3	IDT
LC_St_09_Cy3	Cy3/TTCCGGCTATCCTTTCTGAAGCGGCCCGAGGAGGAAT	IDT
LC_St_14_Cy3	Cy3/TTATAATTTTCAATAGACTTGATAGAACGCTTCTGAGCTATTTGCC	IDT
LC_St_16_Cy3	ACCTGTCGTGCGAAAGCAGGGTCCAGAATGCAGTCCTT/Cy3	IDT
LC_St_20_Cy3	TTATTAATTCTTGCTCTTT/Cy3	IDT

Supplementary Table 1 | Sequences and corresponding modifications of all oligonucleotides. Nomenclature: “FC” stands for a single-stranded DNA (ssDNA) strand that was used to fold FluoroCubes (cube with six dyes). “SD” stands for a ssDNA strand that was used to fold Single Dye Cubes (cube with one dye). “dsDNA” stands for single-stranded DNA (ssDNA) strand that was used to fold the one dye double-stranded DNA (dsDNA). “NH₂” stands for amino group modification. “UN” stands for unlabeled oligonucleotide. “LC” stands for single-stranded DNA (ssDNA) strand that was used to fold the Compact Cube¹.

For each FluoroCube and Single Dye Cube four ssDNA strands are required: SC_01, SC_02, St_01, and St_02. Both Single Dye Cubes (light gray part of the table) and FluoroCubes (white part of the table) need a functionalized oligo strand, which is always St_01 (dark gray part of the table), and either has a biotin or a HALO-ligand². Single Dye Cubes, irrespective of the dye, always require two unlabeled strands “SD_SC_01_UN” and “SD_SC_02_UN” as well as one strand with a single dye “SD_St_02”. Thus, for example, to assemble a Single Dye Cube with ATTO488 and biotin, one uses: SD_SC_01_UN, SD_SC_02_UN, FC_St_01_Biotin, and SD_St_02_ATTO488. For a FluoroCube with biotin one uses: FC_SC_01_ATTO647N, FC_SC_02_ATTO647N, FC_St_01_Biotin, and FC_St_02_ATTO647N. A detailed overview of strand combinations used for Single Dye Cubes, FluoroCubes, one dye double-stranded DNA, and the Compact Cube with one or six dyes can be found in **Supplementary Table 2**. Note that amino group modified oligos were used to label with either JF549 or JF646 (For details, see manuscript on BioRxiv). Here “Cy3” stands for the non-sulfonated version of Cy3 whereas “Cy3N” stands for the sulfonated version of Cy3.

References

1. Scheible, M. B. *et al.* A Compact DNA Cube with Side Length 10 nm. *Small* **11**, 5200–5205 (2015).
2. Los, G. V. *et al.* HaloTag: a novel protein labeling technology for cell imaging and protein analysis. *ACS Chem. Biol.* **3**, 373–382 (2008).

Supplementary Table 2

Fluorocube / Single Dye Cube/ One dye dsDNA/ Compact Cube name	Oligonucleotide strands used
FC_ATTO488_Biotin	FC_SC_01_ATTO488, FC_SC_02_ATTO488, FC_St_01_Biotin, FC_St_02_ATTO488
FC_ATTO565_Biotin	FC_SC_01_ATTO565, FC_SC_02_ATTO565, FC_St_01_Biotin, FC_St_02_ATTO565
FC_ATTO647N_Biotin	FC_SC_01_ATTO647N, FC_SC_02_ATTO647N, FC_St_01_Biotin, FC_St_02_ATTO647N
FC_ATTO647N_HALO	FC_SC_01_ATTO647N, FC_SC_02_ATTO647N, FC_St_01_HALO, FC_St_02_ATTO647N
FC_Cy3_Biotin	FC_SC_01_Cy3, FC_SC_02_Cy3, FC_St_01_Biotin, FC_St_02_Cy3
FC_Cy3N_Biotin	FC_SC_01_Cy3N, FC_SC_02_Cy3N, FC_St_01_Biotin, FC_St_02_Cy3N
FC_Cy5_Biotin	FC_SC_01_Cy5, FC_SC_02_Cy5, FC_St_01_Biotin, FC_St_02_Cy5
FC_JF549_Biotin	FC_SC_01_NH2, FC_SC_02_NH2, FC_St_01_Biotin, FC_St_02_NH2
FC_JF646_Biotin	FC_SC_01_NH2, FC_SC_02_NH2, FC_St_01_Biotin, FC_St_02_NH2
SD_ATTO488_Biotin	SD_SC_01_UN, SD_SC_02_UN, FC_St_01_Biotin, SD_St_02_ATTO488
SD_ATTO565_Biotin	SD_SC_01_UN, SD_SC_02_UN, FC_St_01_Biotin, SD_St_02_ATTO565
SD_ATTO647N_Biotin	SD_SC_01_UN, SD_SC_02_UN, FC_St_01_Biotin, SD_St_02_ATTO647N
SD_Cy3_Biotin	SD_SC_01_UN, SD_SC_02_UN, FC_St_01_Biotin, SD_St_02_Cy3
SD_Cy5_Biotin	SD_SC_01_UN, SD_SC_02_UN, FC_St_01_Biotin, SD_St_02_Cy5
SD_Cy3N_Biotin_SDC1	SD_SC_01_5pr_Cy3N, SD_SC_02_UN, FC_St_01_Biotin, SD_St_02_UN
SD_Cy3N_Biotin_SDC2	SD_SC_01_3pr_Cy3N, SD_SC_02_UN, FC_St_01_Biotin, SD_St_02_UN
SD_Cy3N_Biotin_SDC3	SD_SC_01_UN, SD_SC_02_5pr_Cy3N, FC_St_01_Biotin, SD_St_02_UN
SD_Cy3N_Biotin_SDC4	SD_SC_01_UN, SD_SC_02_3pr_Cy3N, FC_St_01_Biotin, SD_St_02_UN
SD_Cy3N_Biotin_SDC5	SD_SC_01_UN, SD_SC_02_UN, FC_St_01_Biotin, SD_St_02_5pr_Cy3N
SD_Cy3N_Biotin_SDC6	SD_SC_01_UN, SD_SC_02_UN, FC_St_01_Biotin, SD_St_02_3pr_Cy3N
dsDNA_ATTO488_Biotin	dsDNA_Biotin, SD_St_02_ATTO488
dsDNA_ATTO565_Biotin	dsDNA_Biotin, SD_St_02_ATTO565
dsDNA_ATTO647N_Biotin	dsDNA_Biotin, SD_St_02_ATTO647N
dsDNA_Cy3_Biotin	dsDNA_Biotin, SD_St_02_Cy3

dsDNA_Cy5_Biotin	dsDNA_Biotin, SD_St_02_Cy5
dsDNA_Cy3N_Biotin	dsDNA_Biotin, SD_St_02_5pr_Cy3N
LC_one-dye_Cy3_Biotin	LC_SC_01, LC_SC_02, LC_SC_03, LC_SC_04, LC_SC_05, LC_SC_06, LC_SC_07, LC_SC_08, LC_St_01, LC_St_02, LC_St_03, LC_St_04, LC_St_05_Biotin, LC_St_06, LC_St_07_Cy3, LC_St_08, LC_St_09, LC_St_10, LC_St_11, LC_St_12, LC_St_13, LC_St_14, LC_St_15, LC_St_16, LC_St_17, LC_St_18, LC_St_19, LC_St_20
LC_six-dyes_Cy3_Biotin	LC_SC_01, LC_SC_02, LC_SC_03, LC_SC_04, LC_SC_05, LC_SC_06, LC_SC_07, LC_SC_08, LC_St_01, LC_St_02, LC_St_03_Cy3, LC_St_04, LC_St_05_Biotin, LC_St_06, LC_St_07_Cy3, LC_St_08, LC_St_09_Cy3, LC_St_10, LC_St_11, LC_St_12, LC_St_13, LC_St_14_Cy3, LC_St_15, LC_St_16_Cy3, LC_St_17, LC_St_18, LC_St_19, LC_St_20_Cy3

Supplementary Table 2 | Combination of oligonucleotide strands used to assemble all DNA FluoroCubes, Single Dye Cubes, one dye double-stranded DNA and Compact Cubes with one or six dyes. Nomenclature: “FC” stands for FluoroCubes with six dyes and “SD” stands for Single Dye Cubes. “dsDNA” stands for one dye double-stranded DNA (dsDNA). “LC” stands for the Compact Cube¹. The exact sequences with modifications are given in **Supplementary Table 1**. Here “Cy3” stands for the non-sulfonated version of Cy3 whereas “Cy3N” stands for the sulfonated version of Cy3. The detailed design of the Cy3N Single Dye Cubes (SDC 1-6) is shown in the manuscript on BioRxiv. The SDC 5 design of the Cy3N Single Dye Cubes has the dye at the same position as all other Single Dye Cubes with different dyes. Note, that for FluoroCubes with either Janelia Fluorophores JF549 or JF646 amino group modified oligos have been used which were labeled with either dye as described in the **Online Methods** of the manuscript on BioRxiv.

Reference

1. Scheible, M. B. *et al.* A Compact DNA Cube with Side Length 10 nm. *Small* **11**, 5200–5205 (2015).